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Machine Learning in Portfolio Optimization under the SAFE AI Framework

An Empirical Study of Expected Return Estimation

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Better predictions are not enough

Portfolio optimization depends on expected-return estimates — the hardest quantity to estimate reliably. Small errors reshape the entire allocation.

The Historical Mean is simple and transparent, but uses only past returns. Machine learning can exploit a far richer set of stock-level characteristics.

Yet when a model guides real investment decisions, predictive accuracy alone is not enough — we must also ask whether it is robust, fair, explainable and trustworthy.

The promise

Can machine-learning estimates generate better portfolios than the Historical Mean?

The catch

Even if it does — can we trust a model we cannot see inside?

Two questions, rarely asked together

Portfolio research emphasizes predictive or financial performance; Responsible AI emphasizes robustness, fairness and explainability. This thesis connects the two.

1

PERFORMANCE

Does XGBoost generate better out-of-sample portfolio performance than the Historical Mean benchmark, in the same constrained framework?

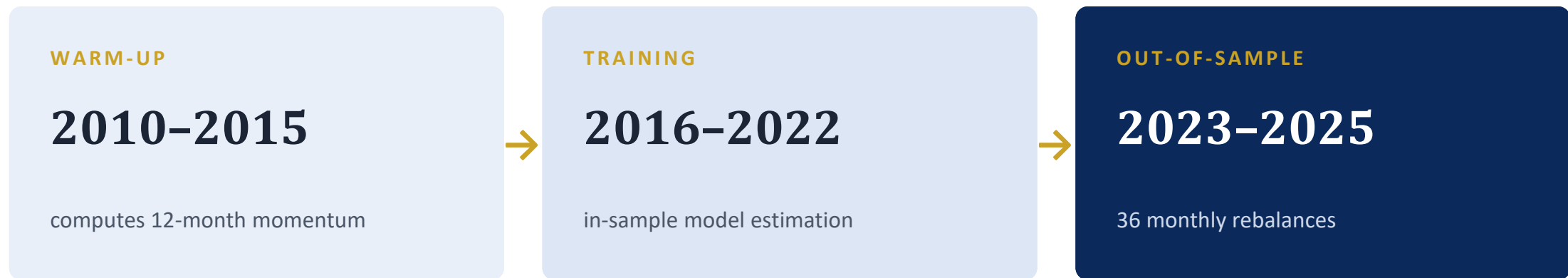
2

TRUSTWORTHINESS

Is the machine-learning model with the highest four-dimensional SAFE AI compliance score also the one that achieves the best risk-adjusted portfolio performance?

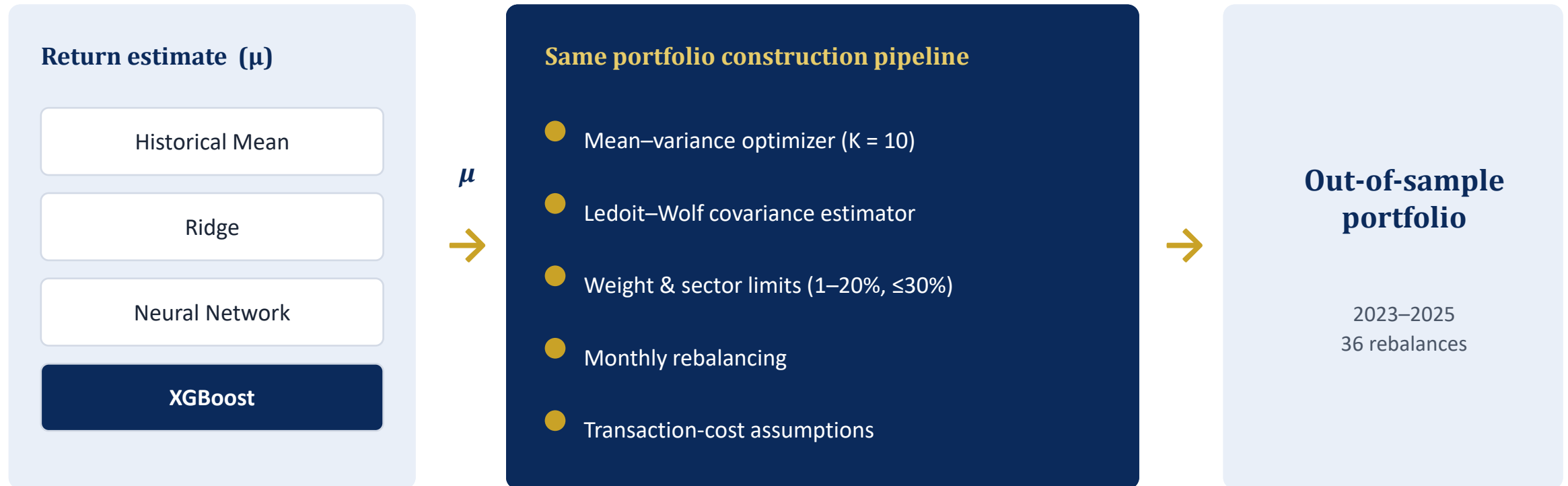
S&P 500, 2010–2025 — a walk-forward out-of-sample design

Bloomberg S&P 500 · 407 stocks after cleaning · **four estimators: Historical Mean, Ridge, Neural Network, XGBoost.**



Walk-forward evaluation. At each rebalance, the machine-learning models are re-estimated on an expanding window, using only past information — no look-ahead.

One pipeline — only the return estimate changes



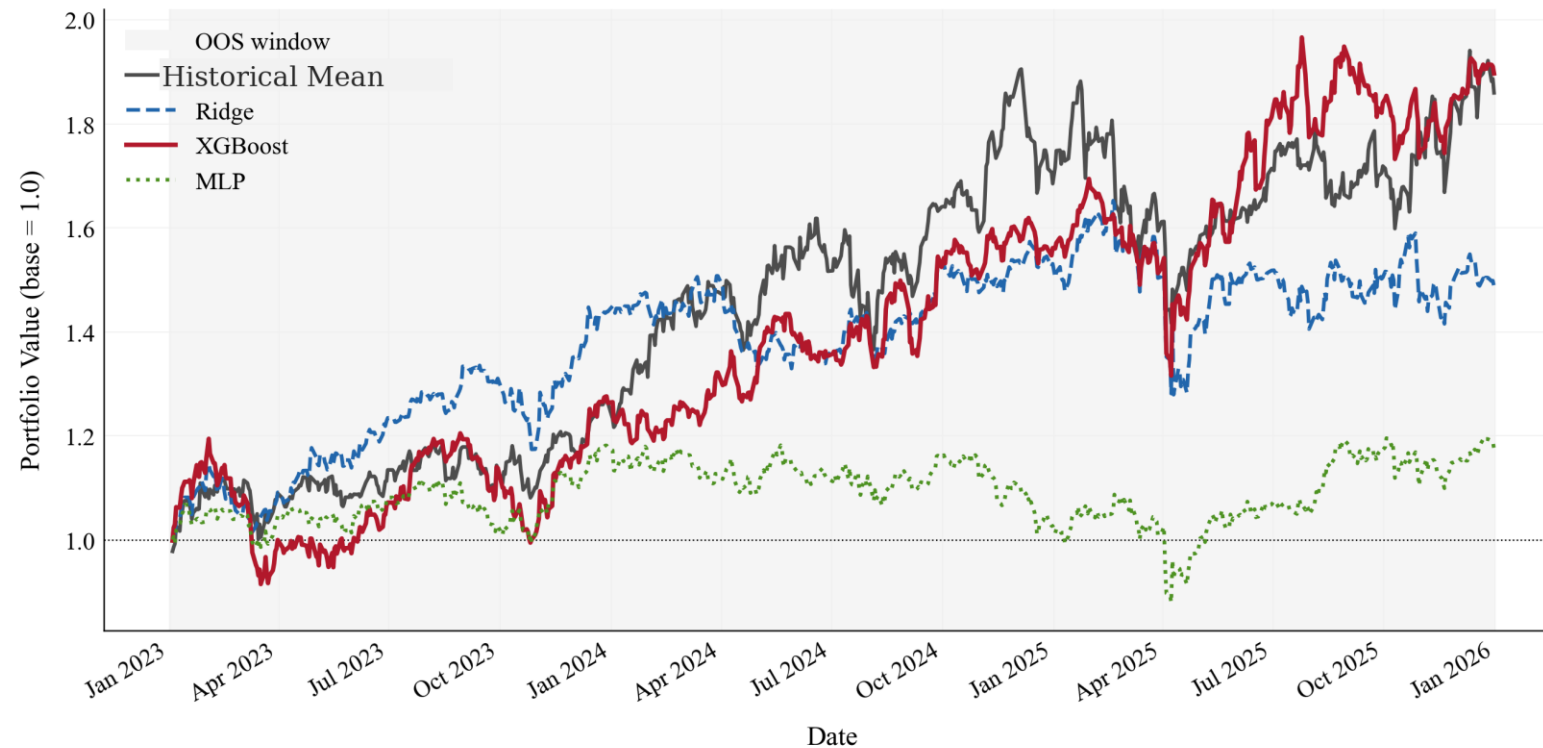
ML models predict cross-sectional ranks, rescaled to the Historical-Mean scale. Because only μ changes, differences in performance are attributable primarily to the estimator.

At $\lambda = 1$, XGBoost delivers the highest risk-adjusted performance

Estimator	Ann. Return	Volatility	Sharpe	Max DD (loss)	Turnover	Net Sharpe (30bps)
XGBoost	21.60%	23.98%	0.90	23.41%	78.2%	0.78
Historical Mean	20.93%	25.39%	0.82	25.66%	46.5%	0.76
Ridge	13.39%	21.76%	0.62	22.58%	79.9%	0.48
Neural Network	5.45%	18.04%	0.30	25.57%	93.2%	0.12

Higher return, lower volatility, and the best Sharpe of the four — a narrow edge survives transaction costs.
Risk–return trade-off check: XGBoost leads at $\lambda = 1$ and 2; the Historical Mean is more competitive at other λ . So the gain is real but conditional.

Cumulative wealth, 2023–2025 (gross, base = 1.0)

**XGBoost**

finishes highest (1.90), but only overtakes the Historical Mean late in the sample.

Historical Mean

stays close (1.86) with substantially lower turnover.

Neural Network (MLP)

low cumulative performance (1.18) and near-total monthly turnover.

XGBoost achieves the highest overall SAFE AI compliance

Scored models: Ridge, Neural Network, XGBoost. *The Historical Mean is a benchmark, not a learning model, so it is not evaluated here.*

Four rank-based dimensions

Robustness · Accuracy · Fairness · Explainability.

Compliance vectors integrated into CS4

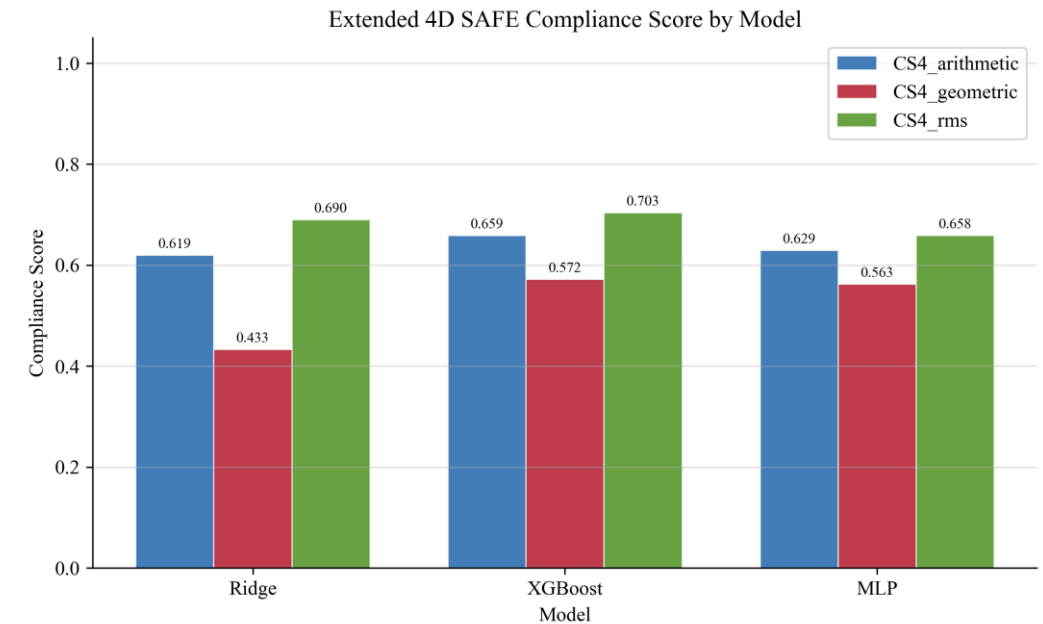
RGR · RGA · RGF* · RGE*.

No model wins every dimension

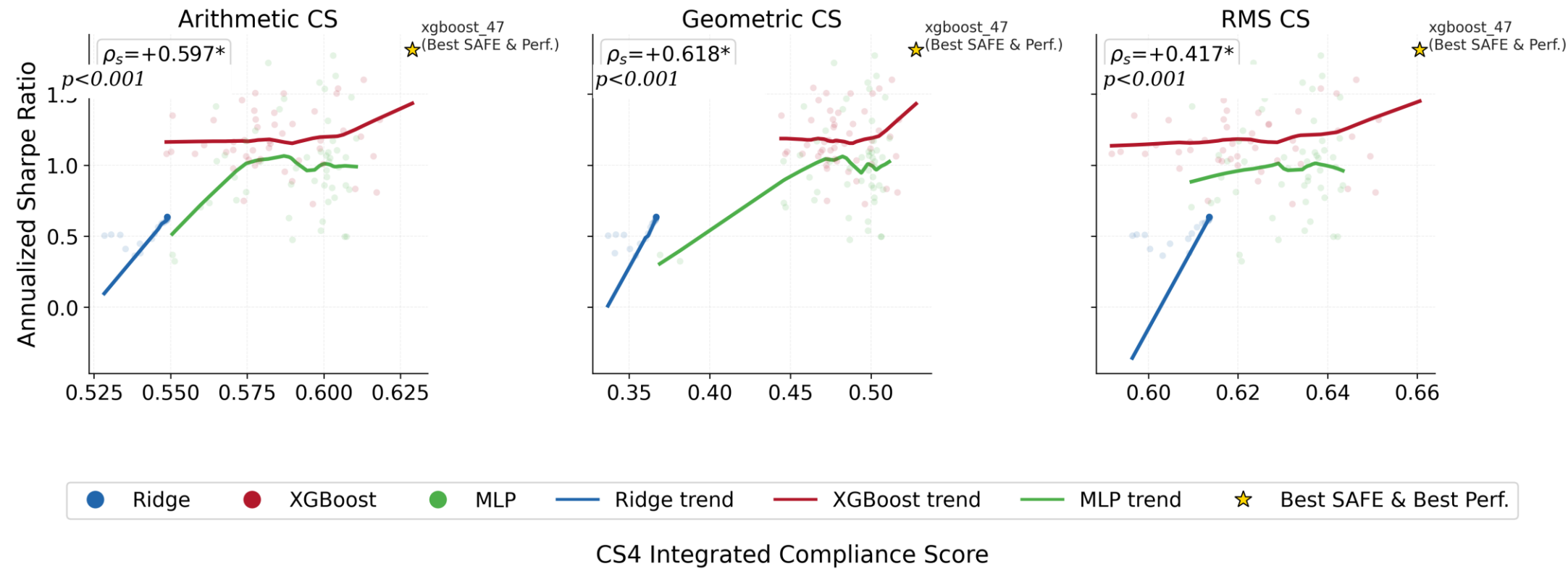
Accuracy & Robustness → XGBoost · Fairness → Neural Network · Explainability → Ridge.

But combined, XGBoost is first under all three aggregations

— arithmetic, geometric and RMS — the strongest overall balance.

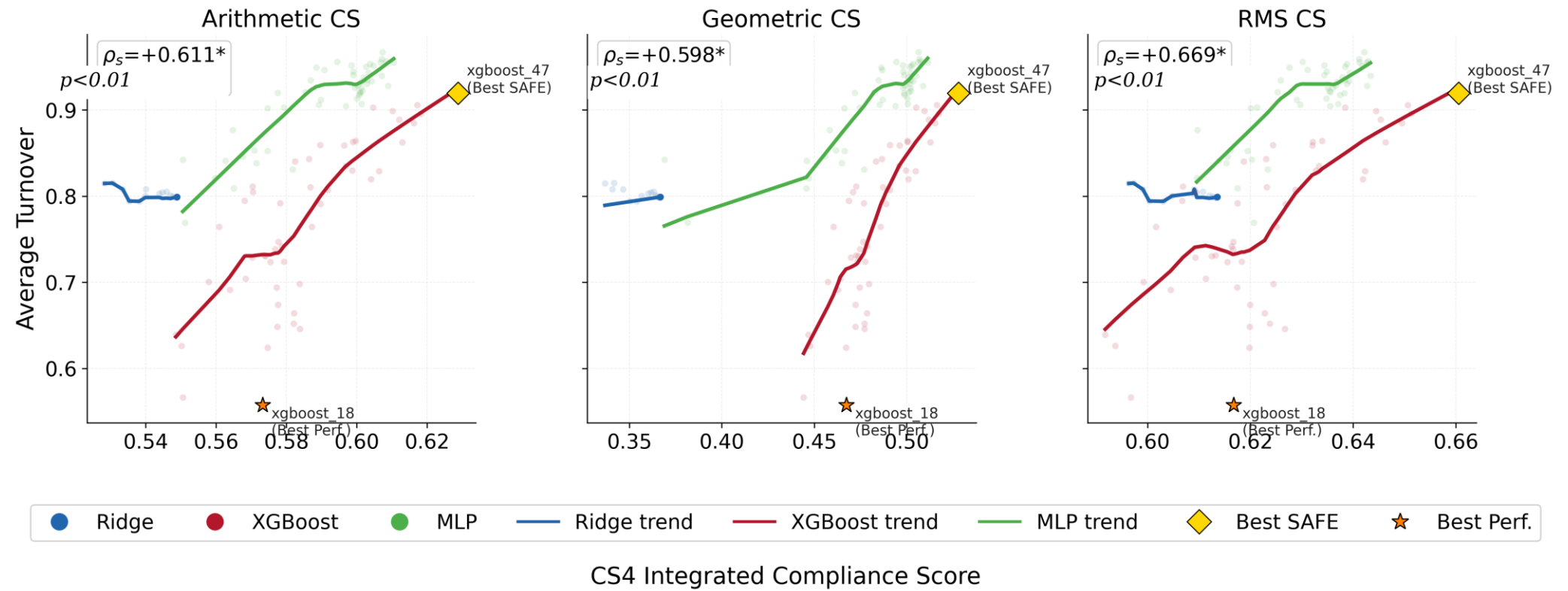


SAFE compliance and performance align — at the aggregate level



$\rho(\text{CS4}, \text{Sharpe}) = 0.597 / 0.618 / 0.417$ (arithmetic / geometric / RMS), all $p < 0.001$. ★ **xgboost_47** — the strongest overall configuration in the Chapter 6 analysis, combining the highest Sharpe with a high CS4.

The implementation trade-off: higher compliance, higher turnover



$\rho(\text{CS4, turnover}) = 0.611 / 0.598 / 0.669$ (arithmetic / geometric / RMS), all $p < 0.01$. Higher compliance also aligns with **lower maximum drawdown** — SAFE AI is a selection criterion, not a performance guarantee.

CONCLUSION

Trustworthy and profitable can align

1

RQ 1 — answered

XGBoost delivers higher risk-adjusted performance in the main setting — conditional on the risk–return trade-off parameter and reduced by transaction costs.

2

RQ 2 — answered

In the main setting, the best-performing model also has the highest CS4.

3

The broader result

At the aggregate level, higher CS4 is associated with higher Sharpe and lower drawdown, though not uniformly within families.

4

The honest limit

The cleaned S&P 500 universe and the short 2023–2025 out-of-sample period limit generalizability.

CONTRIBUTION

Connecting portfolio performance and model SAFE AI compliance in a single empirical framework — the SAFE–Performance Frontier. Portfolio optimization is not only about better predictions; it is about better investment decisions.

Thank you

Questions welcome

Anila Vata · MSc in Finance, Università di Pavia

Supervisor: Prof. Paolo Giudici · 21 July 2026

Does the advantage survive λ and costs?

λ sensitivity — Sharpe ratio

λ	Hist. Mean	XGBoost
0.1	1.28	1.15
0.2	1.20	1.12
0.5	1.02	0.99
1.0	0.82	0.90
2.0	0.82	0.85
5.0	0.94	0.84
10.0	0.95	0.81

XGBoost has the highest Sharpe only at $\lambda = 1$ and $\lambda = 2$; elsewhere the Historical Mean leads. So the result is conditional on the portfolio objective.

Implementation ($\lambda = 1$)

	XGBoost	Hist. Mean
Gross Sharpe	0.90	0.82
Net Sharpe (30 bps)	0.78	0.76
Avg. monthly turnover	78.2%	46.5%

Takeaway. XGBoost keeps a narrow net advantage at $\lambda = 1$, but its higher turnover reduces implementability.

How SAFE AI was operationalized

Dimension	Scalar metric	Compliance vector	Scalar leader
Accuracy	Mean RGA	RGA vector	XGBoost
Robustness	Group RGR at 5%	RGR vector	XGBoost
Fairness	1 – mean sector RGA gap	RGF*	Neural Network
Explainability	Maximum single-feature RGE	RGE*	Ridge

- The Historical Mean is not scored — it is not a trained learning model.
- Fairness concerns groups of securities (GICS sectors), not demographic groups.
- Scalar metrics identify dimensional leaders; the compliance vectors are integrated into CS4.

Ridge has the highest maximum single-feature RGE, mainly associated with firm size.

Statistical evidence behind the frontier

Outcome	Arithmetic	Geometric	RMS	Interpretation
Sharpe ratio	+0.597	+0.618	+0.417	<i>all $p < 0.001$</i>
Max drawdown	−0.35	−0.34	−0.36	<i>all $p < 0.01$</i>
Turnover	+0.611	+0.598	+0.669	<i>all $p < 0.01$</i>

Within-family caveat — arithmetic CS4. The CS4–Sharpe relationship is significant for Ridge ($\rho = 0.498$, $p < 0.001$), but not for XGBoost ($\rho \approx 0.09$, $p = 0.54$) or the Neural Network ($\rho = 0.15$, $p = 0.29$).

Interpretation. Part of the aggregate relationship reflects differences between model families, not within them.